

1. What is the primary function of a router in a computer network?

- a) Assigning IP addresses to devices
- b) Providing wireless connectivity to devices
- c) Forwarding data packets between networks
- d) Managing user authentication and access control

Ans - c) Forwarding data packets between networks

2. What is the purpose of DNS (Domain Name System) in a computer network?

- a) Encrypting data transmissions for security
- b) Assigning IP addresses to devices dynamically
- c) Converting domain names to IP addresses
- d) Routing data

Ans - b) Assigning IP addresses to devices dynamically

3. What type of network topology uses a centralized hub or switch to connect all devices?

- a) Star
- b) Bus
- c) Ring
- d) Mesh

Ans - a) Star

4. Which network protocol is commonly used for securely accessing and transferring files over a network?

- a) HTTP
- b) FTP
- c) SMTP
- d) POP3

Ans - b) FTP

5. True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Ans - true

6. True or False: DHCP (Dynamic Host Configuration Protocol) assigns static IP addresses to network devices automatically.

Ans- False

7. True or False: VLANs (Virtual Local Area Networks) enable network segmentation by dividing a single physical network into multiple logical networks.

Ans- True

8. Explain the difference between a hub and a switch in a computer network.

Ans – Hub

- works on Physical layer of network**
- hub broadcast's data**
- hub work on half duplex**
- hub send's data on binary**

Switch

- Works on data link layer**
- switch multicast data**
- switch work on full duplex**
- switch send's data in frames**

9. Describe the process of troubleshooting network connectivity issues.

Ans- First try to trouble shoot the network of computer

-Check for computer is given proper ip or connected to proper network. And try It to connect by other internet or try by usb tethering mode or restart the pc.

-Check for ping

-if device is connected to network through wired connection so check that wire or port is connected properly

-check the cable is working properly by network cable tester.

-check for router , switch or hub are working properly and not so try It to unplug and restart it .

-still there is issue so call for network support.

11. Discuss the importance of network documentation and provide examples of information that should be documented.

- **Ans- helps fix problems faster. if the internet stops working, you can look at your notes to see what might be wrong.**
- **keeps things organized. you'll know what devices are in your network and where they are connected.**
- **makes upgrades easier. if you want to add a new switch or router, you'll know where to connect it.**
- **good for security. you can keep track of passwords and settings in a safe way.**
- **network diagram. a simple drawing that shows how all the devices are connected (like in star topology).**